# What does this project do?

field-desk: a CRM for a distribution business, extended with two accounting corrections

## In one paragraph

field-desk is a web application that a distribution company uses to run its customer-facing workflow: Sales punches orders, a key-account manager (KAM) approves them, the Warehouse loads, dispatches and confirms deliveries and records returns, and Accounts records customer payments, credit notes and reward points. Every money-affecting step goes through one strict, audited "action" pipeline, so customer balances always add up.

## The people (roles) who use it

| Role | What they do |
|---|---|
| Sales | Create customers, punch and edit orders, cancel unshipped quantity |
| KAM | Approve or reject orders; authorise corrections |
| Warehouse | Start loading, dispatch, confirm delivery, record returns |
| Accounts | Receive payments, allocate cash, issue and apply credit notes, redeem points |
| Admin | Can do everything |

## The money model in plain words

- **Charge:** when goods are delivered, the customer owes money for those goods (one goods charge per delivery confirmation).
- **Receipt:** cash the customer paid. Cash is applied to the oldest outstanding charges first. Cash not yet applied is "free cash".
- **Credit note:** a part of a receipt reserved as credit, which can be applied to a charge or refunded.
- **Reward points:** earned when goods are paid for in cash; locked while unpaid; clawed back if the payment is reversed after the points were spent.
- **Append-only history:** nothing is edited or deleted. A mistake is fixed by adding a compensating (opposite) entry.

## What I added (the take-home task)

The assignment ("Business Rules Engine") asked for two real features while keeping the code from turning into a tangle of special cases.

| Ticket | Problem | What the system now does |
|---|---|---|
| FDSK-169 | Accounts recorded a payment against the wrong customer. | Command **MOVE_PAYMENT**: the receipt is marked reversed, an identical receipt is created on the right customer, and both customers' charges, cash and points are recalculated. |

| Ticket | Problem | What the system now does |
|---|---|---|
| FDSK-170 | A KAM marked 10 units delivered but only 6 arrived. | Command **CORRECT_DELIVERY**: delivery is lowered to 6 using reversal rows, the charge shrinks, extra cash is released, points are re-locked. When the other 4 are confirmed later, everything re-applies automatically. |

## Technology

- Monorepo with pnpm workspaces: **client** (React + Vite), **server** (NestJS + Drizzle ORM + SQLite), **shared** (Zod contracts used by both).
- Tests with the Node test runner against a real SQLite database through HTTP.
- Private repo Aman-0402/field-desk, imported from AviKKi/field-desk with its original history kept.